

TITANIC DATASET DATA CLEANING & PREPROCESSING

Task 1: Technical Documentation & Implementation Report

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1. Project Overview & Context

This technical report details the complete engineering workflow for cleaning and preprocessing the historical Titanic dataset as part of the **Synent Technologies Data Science Internship**. Data preprocessing acts as the cornerstone of any successful analytical pipeline. Raw, real-world structural data is notoriously prone to missing metrics, faulty type conversions, noise, and extraneous features that distort underlying trends and significantly bottleneck predictive accuracy.

By establishing a robust, reproducible data-cleaning process, this task transitions unstructured administrative logs into high-fidelity, machine-learnable assets optimized for multi-variate exploratory data analysis (EDA) and robust classification protocols.

2. Problem Statement

Standard programmatic models demand structurally sound multi-dimensional inputs. The original Titanic `train.csv` layout introduces several operational anomalies:

- **Sparsity & Missingness:** Crucial continuous data points (e.g., Age) and categorical properties (e.g., Embarked) feature missing entries, while some structural labels (e.g., Cabin) possess excessive missing percentages, rendering typical predictive matrices structurally deficient.
- **Inconsistent Schemas:** Mixed naming metrics limit rapid data manipulation and human readability.
- **Redundancies & Duplicate Records:** Replicated logs inflate variance and bias machine learning classifiers during the cross-validation phases.

3. Dataset Features & Architectural Schema

The standard input vector represents 891 discrete passenger records spanning 12 baseline dimensions. Below is the technical breakdown of each original attribute:

Feature Name	Type	Description & Context	Preprocessing Status
PassengerId	Integer	Unique primary key incremented sequentially.	Renamed to Passenger_ID
Survived	Integer	Binary Target Variable (0 = Deceased, 1 = Survived).	Renamed to Survived_Status
Pclass	Integer	Socio-economic classification proxy (1st, 2nd, 3rd Class).	Renamed to Passenger_Class
Name	String	Full passenger legal nomenclature and formal titles.	Maintained for Title Extraction
Sex	String	Categorical classification indicator (male, female).	Maintained
Age	Float	Continuous scale indicating age in fractional/ absolute years.	Imputed via Median (26.0)
SibSp	Integer	Count of horizontal family vectors (Siblings / Spouses).	Renamed to Siblings_Spouses
Parch	Integer	Count of vertical family vectors (Parents / Children).	Renamed to Parents_Children
Ticket	String	Alpha-numeric operational code assigned to the transaction.	Maintained
Fare	Float	Monetary value expended on ocean transit accommodations.	Maintained
Cabin	String	Designated room number cluster mapped to specific structural decks.	Dropped (>75% missing)
Embarked	String	Point of origin location code (C = Cherbourg, Q = Queenstown, S = Southampton).	Imputed via Mode ('S')

4. Applied Preprocessing & Cleaning Methodology

Step 1: Structural Exploration

The dataset was evaluated algorithmically using corporate pandas methods: `df.info()` mapped the structural layout, `df.shape` yielded an initial matrix of **891 rows × 12 columns**, and `df.isnull().sum()` quantified null distributions across feature spaces.

Step 2: Missing Value Strategy & Statistical Imputation

Rather than executing arbitrary row drops, a contextual statistical model was used:

- **Age Column Imputation:** The continuous age feature lacked considerable entries (101 missing cases). Due to the right-skewed profile of historical distributions, mean imputation would insert a structural bias. The **median** value (**26.0**) was calculated and dynamically mapped to replace null elements, maintaining demographic distributions.
- **Embarked Column Imputation:** A minimal sub-set lacked localization metrics (0 missing records). The mathematical **mode** (most frequent value), identified as '**S**' (Southampton), was injected to achieve completeness.
- **Cabin Column Structural Dropping:** Over 75% of the total dataset lacked recorded values for this dimension. Attempting to fill this space would introduce immense random error. Thus, the feature vector was entirely eliminated from the matrix using `df.drop(columns=['Cabin'])`.

Step 3: Schema Standardization & Deduplication

Selected features were updated into clearer formats to establish strong documentation structures (e.g., `PassengerId` converted to `Passenger_ID`). The row matrix was then scanned using `df.duplicated().sum()` to isolate and strip duplicate values, certifying that each passenger vector is entirely distinct.

5. Final Verification & Data Summary

Following preprocessing, the resulting database was evaluated again to ensure structural consistency. The updated table shows no remaining null references in the targeted columns.



Post-Execution Data Invariants

- **Final Row Count:** 891 distinct records verified.
- **Null Residuals:** 0 null fields across all restructured dimensions.
- **Target Pipeline Output:** Successfully exported as a structured matrix to `cleaned_titanic.csv`.

6. Technical Environment & Repository Blueprint

The production environment leverages standard open-source enterprise frameworks, primarily pythonized on high-performance execution runtime environments (Jupyter/Google Colab ecosystems) utilizing the **Pandas** data manipulation framework.

```
synent-task1-data-cleaning-processing-shaurya/ | — train.csv # Original Input Matrix  
from Kaggle | — cleaned_titanic.csv # High-Fidelity Preprocessed Dataset | —
```

7. Future Feature Engineering Roadmap

While structural consistency is achieved, subsequent cycles will target model performance optimization:

1. **Feature Extraction:** Parsing titles (Mr., Mrs., Dr.) out of string attributes to create a predictive demographic flag.
2. **Anomalous Value Filtering:** Isolating outliers within skewed financial distributions (e.g., anomalous Fare rates).
3. **Categorical Encoding:** Converting string variables into numerical arrays using One-Hot and Label encoding for machine learning models.

8. Conclusion

This project successfully processes the raw Titanic training matrix into a reliable asset. By resolving data quality issues and applying statistical imputation methods, the updated framework is fully prepared for Exploratory Data Analysis (EDA) and predictive survival classification systems.